Perspectives on Privacy-Preserving Information Retrieval*

Extended Abstract Paper

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Abstract

Recent findings in computer science have intensified concerns regarding the privacy of user data submitted to and processed by Information Retrieval (IR) systems. Modern search engines increasingly incorporate generative information access pipelines that couple document retrieval with answer generation, thus raising critical questions for Privacy-Preserving IR (PPIR), e.g., whether such data leak personal information or are employed to train generative models and whether queries with retrieved documents remain protected throughout the retrieval process. We propose a PPIR research agenda organised around novel Privacy Research Pillars (PRPs), whose objectives are grounded in gaps identified through structured interviews with experts, a user survey, and empirical evaluations. These pillars are intended to guide the IR community toward more resilient privacy-preserving IR.

Keywords

Privacy Preserving Information Retrieval, Perspectives in Information Retrieval, Information Security, User Safety

1. Introduction

Information Retrieval (IR) systems, specifically search engines, process increasingly large volumes of data everyday. Queries, clicked documents, and search histories reveal information needs regarding the users' personal sphere, such as health and finance [2, 3, 4, 5], and practices such as *ego-surfing* expose identity attributes directly to the system [6]. The collection, processing, and retention of such data expose users to several risks, e.g., exfiltration of search logs may disclose personal details [7], insiders may exploit query logs for illegitimate purposes, and users may be profiled without consent. In addition to all of this, the integration of Large Language Models (LLMs) into modern search engines [8] and recommender systems further increases these risks, since training on query logs, prompts, and retrieved relevant documents increases the likelihood of memorisation and leakage of information [9, 10].

Privacy-Preserving Information Retrieval (PPIR) addresses these concerns by limiting the disclosure of personal information to both internal and external adversaries during the retrieval, typically at a non-negligible cost in effectiveness or efficiency, e.g., query obfuscation [11, 12, 13, 14, 15, 16, 17], that preserves efficiency but degrades retrieval effectiveness. Also, the evaluation of the privacy afforded by these techniques for unstructured texts remains limited and is generally reduced to the formal parameters analysis of the protection mechanism, such as the budget ε or the encryption key size.

In this paper, we argue that the community requires a common reference framework for developing, evaluating, and comparing PPIR methods. To this end, we put the existing literature into perspective, identify the critical limitations of current solutions, and report the findings of a user study on how experts and general users perceive privacy and IR. Guided by these findings, we define the research objectives and Privacy Research Pillars (PRPs) to drive the future research towards open challenges and relevant issues not not solved in the literature. We refer to the original paper [1] for all the details .

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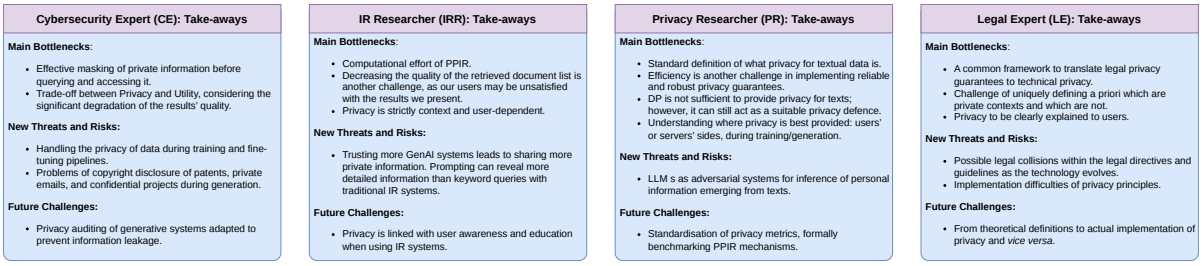


Figure 1: Takeaways from the interviews with experts. The points raised by the experts are organised around the main topics of the interview protocol: bottlenecks in PPIR, new threats and risks, and future challenges.

2. Need for a PPIR Reasearch Agenda

Different definitions of privacy have been proposed through different legal and technical frameworks. In cybersecurity, the National Institute of Standards and Technology Special Pubblication, i.e., NIST SP 800-130 [18], defines "Privacy" as the property of a system to guarantee the "[...] *assurance that the confidentiality of, and access to, certain information about an entity is protected.*". The publication emphasises privacy as an operational characteristic of secure and reliable information systems, focusing on mechanisms necessary to safeguard information. On a more technical level, ε-Differential Privacy (DP), introduced by Dwork et al. [19], provides the *gold standard* mathematical framework for quantifying privacy guarantees when querying and analysing data. Formally, the framework states that, considering a randomisation mechanism $\mathscr{M}$, i.e., an algorithm that takes an input and produces an output, is said to be ε-DP if and only if, for a privacy budget $\varepsilon > 0$ and two neighbouring datasets, i.e., datasets that diverge for at most one entry, D, D', it holds the inequality $\Pr[\mathscr{M}(D) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathscr{M}(D') \in S]$ for all possible outputs $S \subseteq \text{Range}(\mathscr{M})$. The privacy budget ε controls the privacy loss: a smaller ε corresponds to stronger privacy guarantees, and a higher ε provides weaker privacy guarantees. From a human rights perspective, the Universal Declaration of Human Rights [20] establishes privacy as a fundamental right in Article 12, and, in addition, different technology regulations, such as the European Union's GDPR [21], mandate different principles for the handling and processing of personal data [22, 23, 24].

2.1. Experts' and Users' Identified Issues on PPIR

To identify the main objectives for future PPIR research, we draw on the perspectives of experts who work on privacy-related issues. We describe the research challenges and future directions, as outlined by the privacy experts during their interviews. Figure 1 displays boxes that summarise the main findings from such interviews. We developed a structured expert interview protocol comprising open-ended questions to capture the perspectives of four specialists across three domains of privacy implementation: Technical Research, Industry Applications, and Policy and Regulation. Two of the participants were computer science researchers: one focused on privacy in NLP and the other on privacy in IR, both of whom had published extensively on privacy at major computer science conferences. Further, two participants were an industry Cybersecurity Expert and a Legal Expert. We refer to these experts as follows: Privacy Researcher (PR), Information Retrieval Researcher (IRR), Cybersecurity Expert (CE), Legal Expert (LE). The interview protocol is organised into the following key topical sections: i) Main advantages and bottlenecks of current PPIR solutions, ii) Threats and Risks associated with artificial intelligence and privacy when retrieving information, and iii) Future challenges regarding PPIR. The complete interview protocol, including the questions asked, is publicly available in the repository[1]. Each of the four expert interviews was conducted independently by one author.

Following discussions with experts, two primary patterns have emerged: first, there are challenges in defining privacy in unstructured text; second, the evolution of privacy threats is outpacing the development of mitigation strategies. The IR researcher characterises the issue as context- and user-dependent, while the PR emphasises the lack of a standardised definition of private textual data.

[1] https://github.com/Kekkodf/PrivacyResearchAgenda4PPIR.

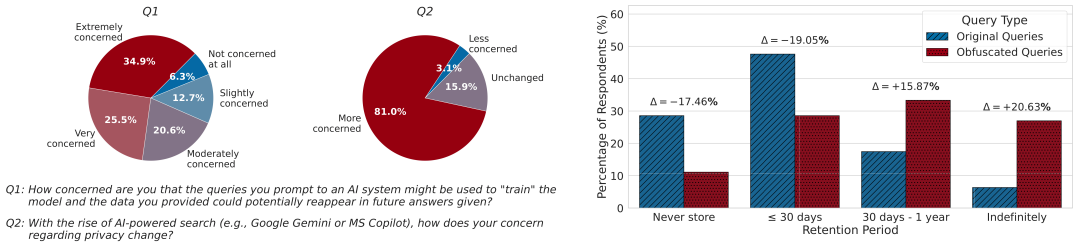


Figure 2: Q1 and Q2 show the results about the search tasks. The histograms report the results on *"If a search engine stores your original/obfuscated queries to improve services, what do you consider an acceptable retention time period for them?"*. The Δ is the % difference for the acceptable retention time of obfuscated *vs.* original queries.

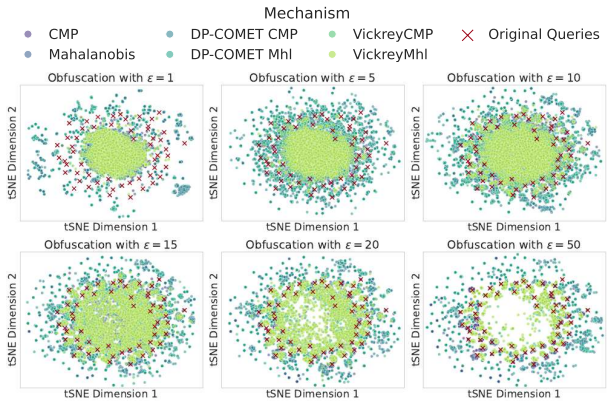


Figure 3: 2D t-SNE of original and obfuscated versions of the TREC Deep Learning 2020 queries [25]. The × are the embeddings of the original queries, while the coloured circles are the ones produced by different obfuscations.

Meanwhile, the LE underscores the difficulty of saying a priori which contexts should be classified as private. Without a standard accepted unit of protection, it is impossible to accurately quantify data leakage. Consequently, standardised metrics and formal benchmarking of PPIR mechanisms are seen as future guarantees rather than established solutions. This situation further complicates the relationship between legal frameworks and technical solutions, deepening the translation issues between legal guarantees and their technical counterparts as expressed by CE and LE.

Figure 2 reports users' privacy attitudes towards AI-powered search and their tolerance of query retention. Concern about prompt data being used by model training and present in generated answers is widespread, with 60.4% of respondents very or extremely concerned and only 6.3% not concerned at all (Q1); moreover, 81.0% state that the diffusion of AI-powered search assistants has increased their concern, against 3.1% reporting a decrease (Q2). Under obfuscation, acceptance shifts systematically from restrictive to permissive retention regimes: the two shortest options decrease by $\Delta = -17.46\%$ and $\Delta = -19.05\%$, while the two longest increase by $\Delta = +15.87\%$ and $\Delta = +20.63\%$, with tolerance of indefinite storage ranging from 6.3% to 27.0%. The symmetry of these deltas suggests a redistribution of the population across the retention scale, supporting the idea that obfuscation serves as a trust-conferring mechanism that relaxes the retention time constraints users would otherwise impose.

2.2. Experimental Evidence on PPIR

To assess such a risk, we use the `all-MiniLM` model [26] to embed the original TREC Deep Learning 2020 [25] queries and their obfuscated versions into a 768-dimensional embedding space. To generate the obfuscated queries, we use the pyPANTERA framework [27]. Such obfuscations are constructed using different DP mechanisms, namely *Noisy Embedding* and *Sampling* [27]. Traditional privacy evaluation measures compute similarity between dense representation vectors to assess the semantics of the underlying texts. Formally, this means computing the cosine of the angle between two vectors in an Euclidean space. To illustrate this, we use t-distributed Stochastic Neighbour Embedding (t-SNE) di-

Table 1
Overviews of Privacy Research Pillars (PRPs) and the respective objectives for a robust development of PPIR.

PRP	Pillar Overview	Objectives addressed	Objectives
Technical Privacy Definitions	Development of new PPIR techniques that better reconcile "formal" and "practical" privacy guarantees for IR.	***O1, O3, O4, O5***	**O1**: Define new privacy solutions for text and unstructured information. **O2**: Develop techniques and metrics to assess and evaluate textual privacy. **O3**: Consider subjective definitions of user privacy and perceptions in PPIR. **O4**: Develop a cohesive framework to translate legal privacy concepts into technical solutions. **O5**: Improve the utility-efficiency-privacy trade-offs of PPIR. **O6**: Improve PPIR resilience for GenAI.
Implementation Rules for PPIR	Implementation guidelines with *privacy-by-design* perspective as core component.	***O3, O4, O5***	
Standard Privacy Evaluation	Outcome aimed at defining a standard way of evaluating the privacy of textual and user information exchanged.	***O2, O3, O6***	

mensionality reduction to obtain a two-dimensional representation of the queries and their obfuscations. Figure 3 shows the t-SNE of the DP query obfuscations at different levels of privacy budget ε. From the results we obtained for the query obfuscation protocol [28, 29], we showed how the embeddings of the obfuscations are distributed in a lower-dimensional vector space relative to the originals. As ε increases, we report a clear trend for almost every DP mechanism: the embeddings of the obfuscated queries converge significantly towards those of the original ones. This geometric proximity implies that the obfuscated variants retain a semantic representation close to the sensitive original one, which is intended to be masked. Thus, the cosine similarity computed using the traditional privacy metric for high ε indicates that the obfuscation mechanisms fail to properly assess the privacy protection for the users' intents. In the real-world an adversary could exploit this high semantic similarity to easily invert the obfuscation and guess the original users' query or use LLMs to assess the queries' similarity [30, 31].

3. Objectives of PPIR and Privacy Research Pillars (PRPs)

Table 1 summarises the three PRPs we propose and their mapping onto the six objectives derived from our expert interviews, user survey, and experiments. The first pillar, *Technical Privacy Definitions*, calls for PPIR techniques that reconcile formal guarantees with the practical protection users expect, addressing **O1**, **O3**, **O4**, and **O5**. The second, *Implementation Rules for PPIR*, asks for *privacy-by-design* guidelines that treat privacy as a core information component and provide a framework for translating legal requirements into technical measures (**O3**, **O4**, **O5**). The third, *Standard Privacy Evaluation*, targets the absence of a shared methodology for assessing the privacy of textual and user information exchanged with IR systems (**O2**, **O3**, **O6**). The mapping is intentionally many-to-many: no single pillar is sufficient in isolation, and **O3** recurs across all three, reflecting our position that subjective notions of privacy must inform the design, the implementation, and the evaluation of PPIR solutions alike.

4. Conclusion

The proposed pillars aim to open new research avenues to develop solutions that address the identified and discussed objectives, thereby mitigating threats and emerging privacy risks in IR. The Privacy Research Pillars (PRPs)' objectives are supported by privacy experts from industry, research, and legislative backgrounds who were interviewed, by users' perspectives collected via an online survey, and by experimental evidence obtained employing state-of-the-art protocols. We claim that these findings provide a robust starting point for developing privacy solutions able to face the confidentiality risks as new systems are released and users interactions with such systems change.

Declaration on Generative AI

During the preparation of this work, the authors used Grammarly for Readability and Spelling checks. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

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